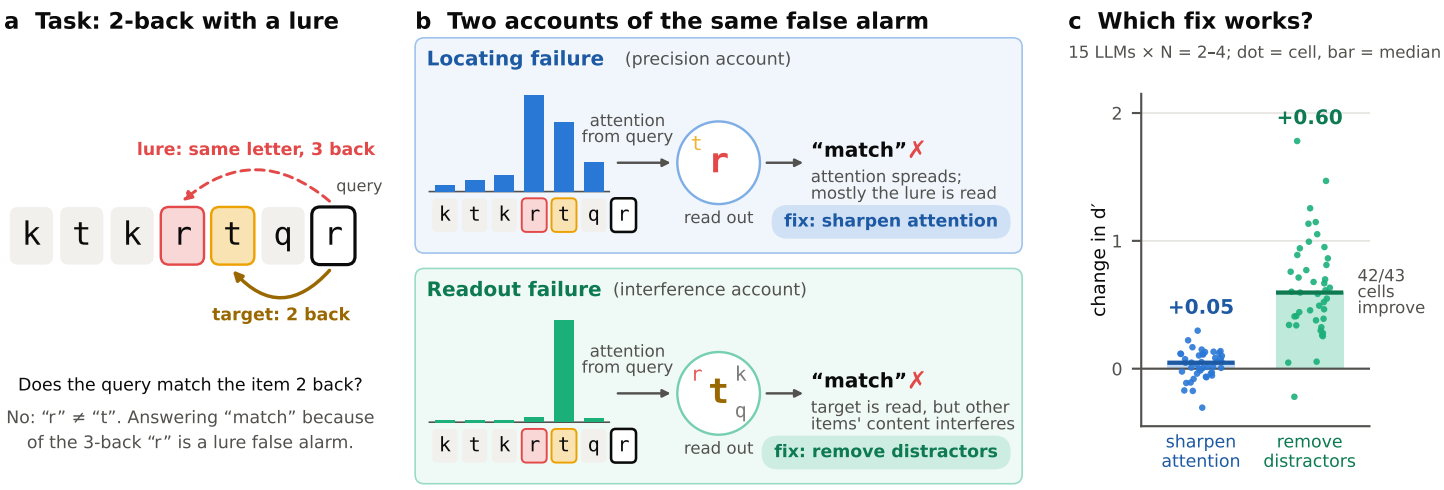
a Task: 2-back with a lure
lure: same letter, 3 back
query
k t k r t q r
target: 2 back
Does the query match the item 2 back?
No: “r” ≠ “t”. Answering “match” because of the 3-back “r” is a lure false alarm.
b Two accounts of the same false alarm
Locating failure (precision account)
attention from query
k t k r t q r
t r
read out
“match” ✗
attention spreads; mostly the lure is read
fix: sharpen attention
Readout failure (interference account)
attention from query
k t k r t q r
r t k q
read out
“match” ✗
target is read, but other items' content interferes
fix: remove distractors
c Which fix works?
15 LLMs × N = 2–4; dot = cell, bar = median
change in d′
2
1
0
+0.05
+0.60
42/43 cells improve
sharpen attention
remove distractors